

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

THERMODYNAMICS

LECTURE-01

PHYSICS

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Topics To Be Covered

- 1** Meaning of Thermodynamics ✓
- 2** Thermodynamics System ✓
- 3** Surrounding ✓
- 4** Type of System ✓
- 5** Extensive Property ✓

Thermodynamics

JK-Board



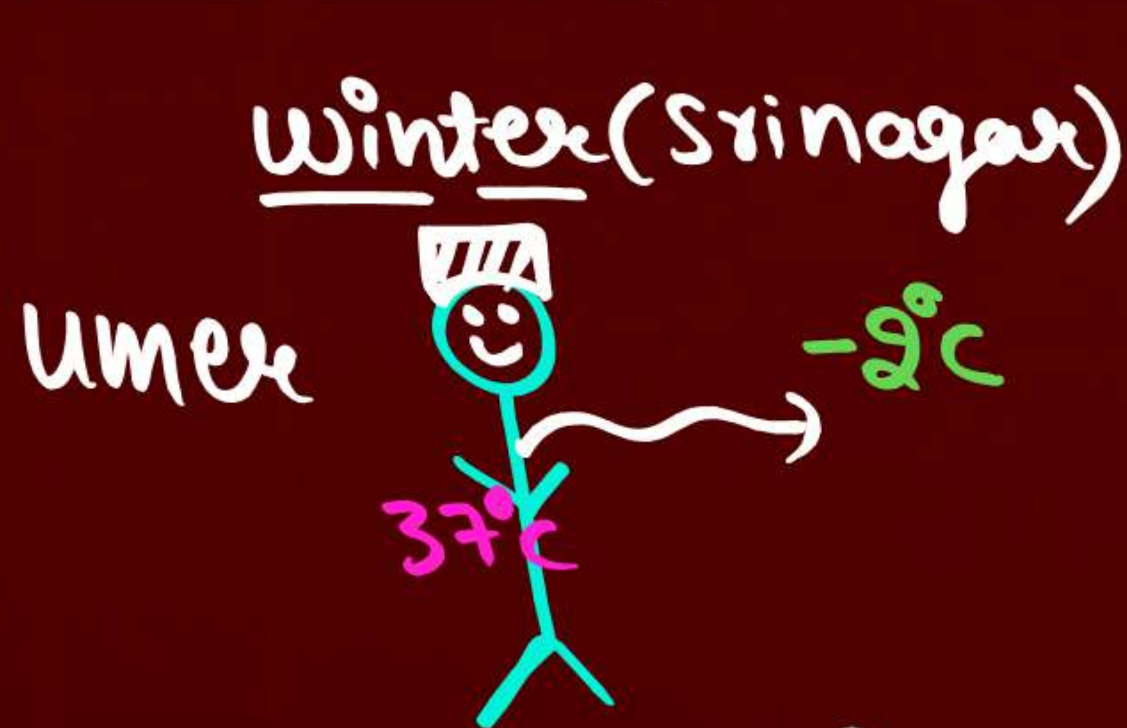
Thermal equilibrium and definition of temperature, zeroth law of thermodynamics, heat, work and internal energy. First law of thermodynamics, Second law of thermodynamics: Gaseous state of matter. change of condition of gaseous state-isothermal, adiabatic, reversible, irreversible, and cyclic processes.

Thermodynamic

Heat flow

①

⇒ The Branch of Physics which deal with the flow of heat



Note :- Heat flow from high temp to low temp



Basic Terms



System

→ The part of universe, on which we investigate

②



Surrounding

→ The Rest of things in universe

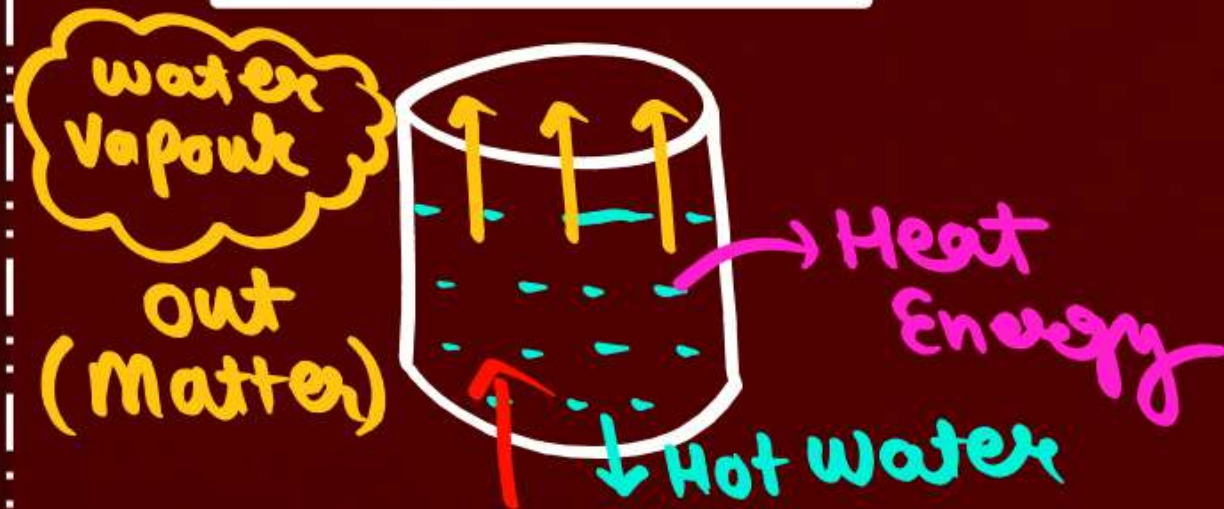
universe = System + Surrounding



System

System

Open System



⇒ The System in which matter and heat Exchange with Surrounding

Close System



⇒ Only heat Energy is Exchange with Surrounding

Isolated System



⇒ heat and matter Both are not Exchange with Surrounding



Thermodynamics 3

(a) Extrinsic Property

→ 2026 (Tammu)
2024 (J & K)



⇒ The property of matter which depend on
Quantity

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mass	1 kg ↓	10 kg ↑
Volume	1 lt ↓	10 lt ↑
Heat	1 J ↓	10 J ↑



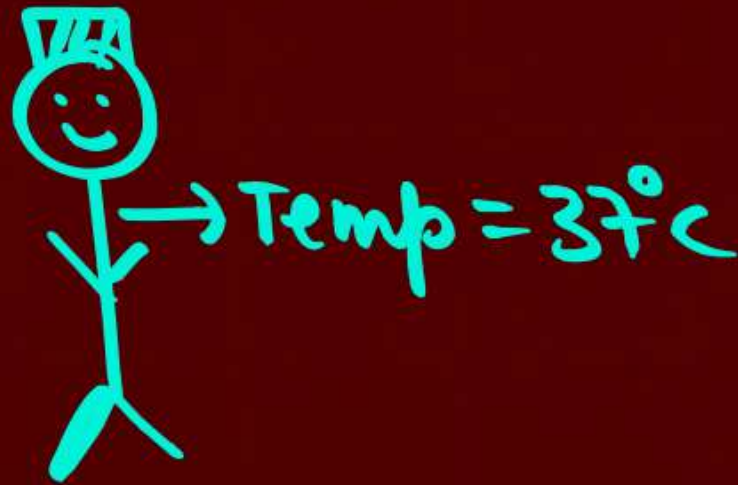
(b) Intrinsic Property

2026 (Jammu) 1M✓

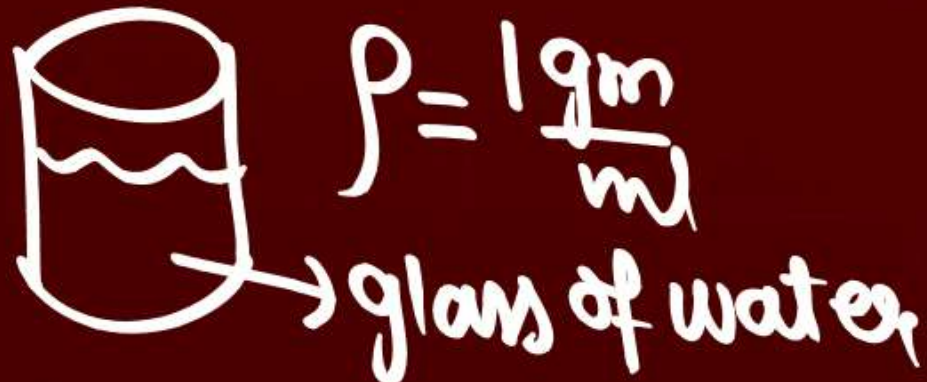


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The property of Matter which donot depend on Quantity (Density ρ and temp)



Temp = 37°C



$\rho = 1 \frac{\text{gm}}{\text{ml}}$

glass of water

Temp = 37°C

Water $\rho = 1 \frac{\text{gm}}{\text{ml}}$



Process

2023

2024

2026



(1) Isothermal Process

⇒ The process in which
temperature remain
Constant

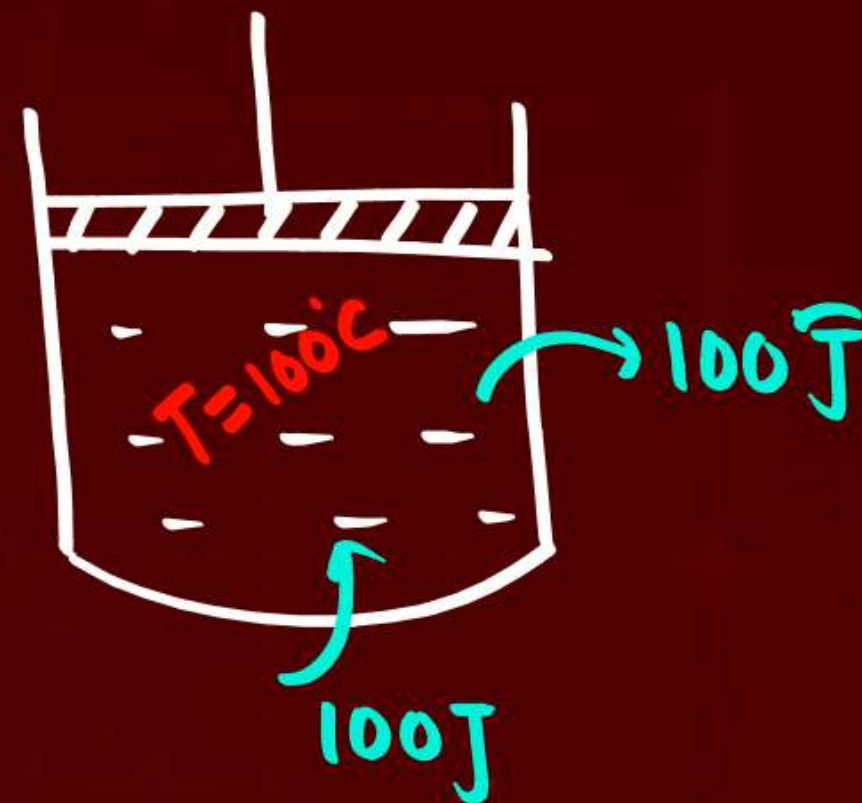
$$T_i = 100^\circ\text{C} \quad T_f = 100^\circ\text{C}$$

Change $\Delta T = T_f - T_i$

$$= 100 - 100 = 0$$

$$\Delta T = 0$$

$$dT = 0 \quad \checkmark$$



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(II) Isobaric process

Same

The process in which
pressure Remain
Constant (c)

$$P_i = 50 \text{ bar}$$

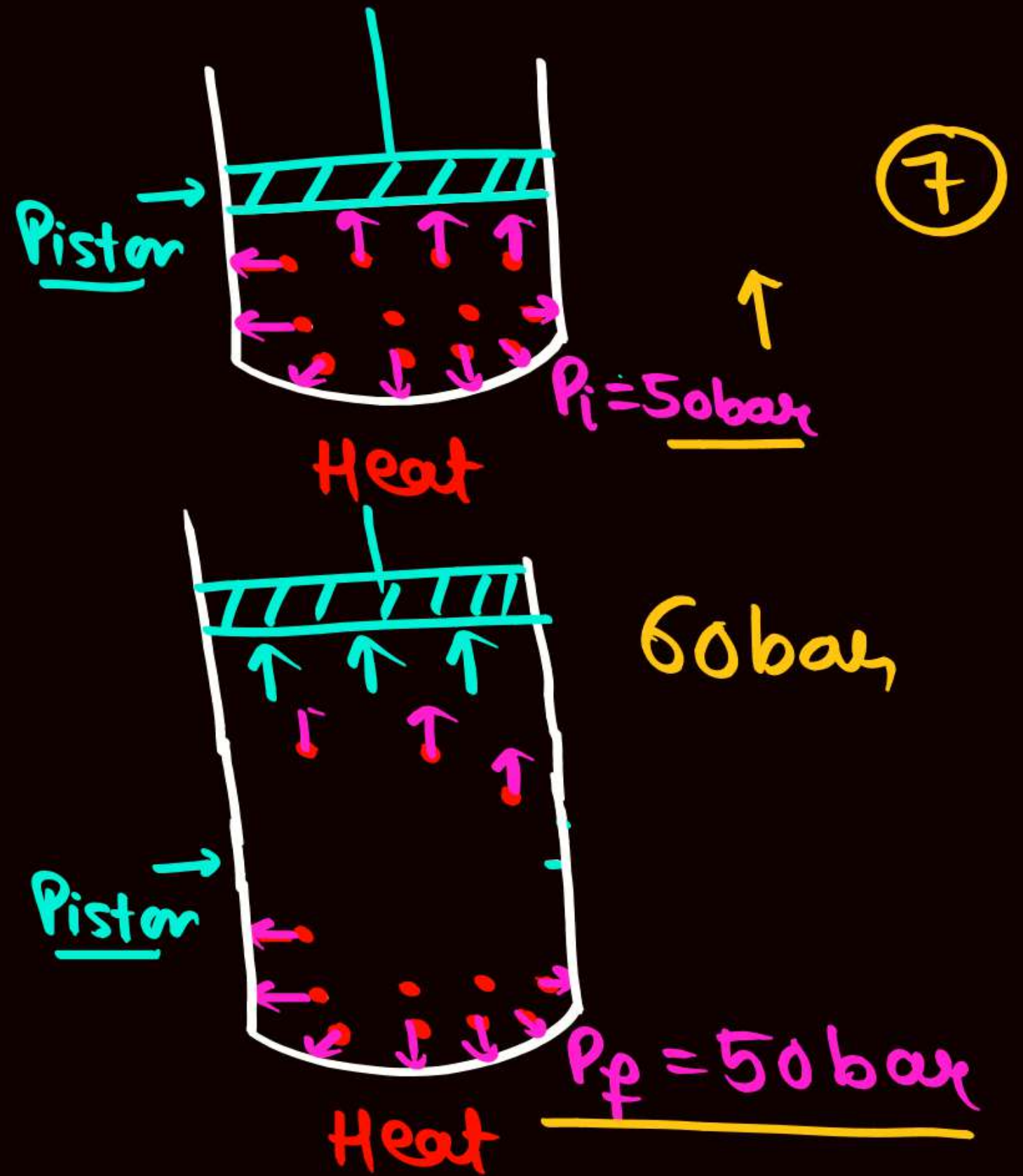
$$P_f = 50 \text{ bar}$$

$$\Delta P = P_f - P_i$$

$$\Delta P = 50 - 50 = \underline{0}$$

$$\Delta P = 0$$

$$dP = 0$$



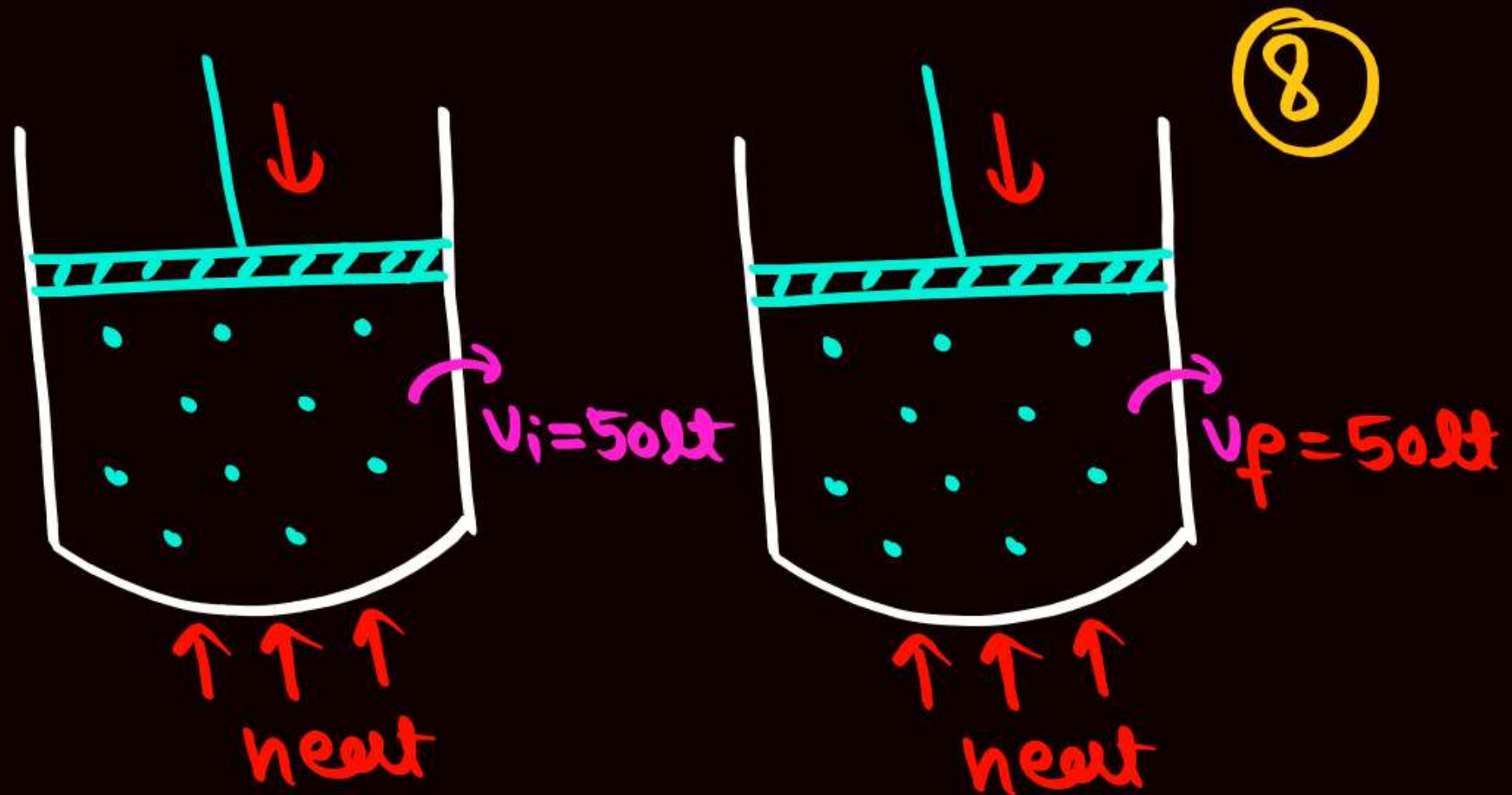
III Isochoric Process

⇒ The process in which
Volume Remain Same

$$V_i = 50 \text{ lt} \quad V_f = 50 \text{ lt}$$

$$\begin{aligned}\Delta V &= V_f - V_i \\ &= 50 - 50 \\ &= 0\end{aligned}$$

$$\begin{aligned}\Delta V &= 0 \\ \underline{dV} &= 0.\end{aligned}$$



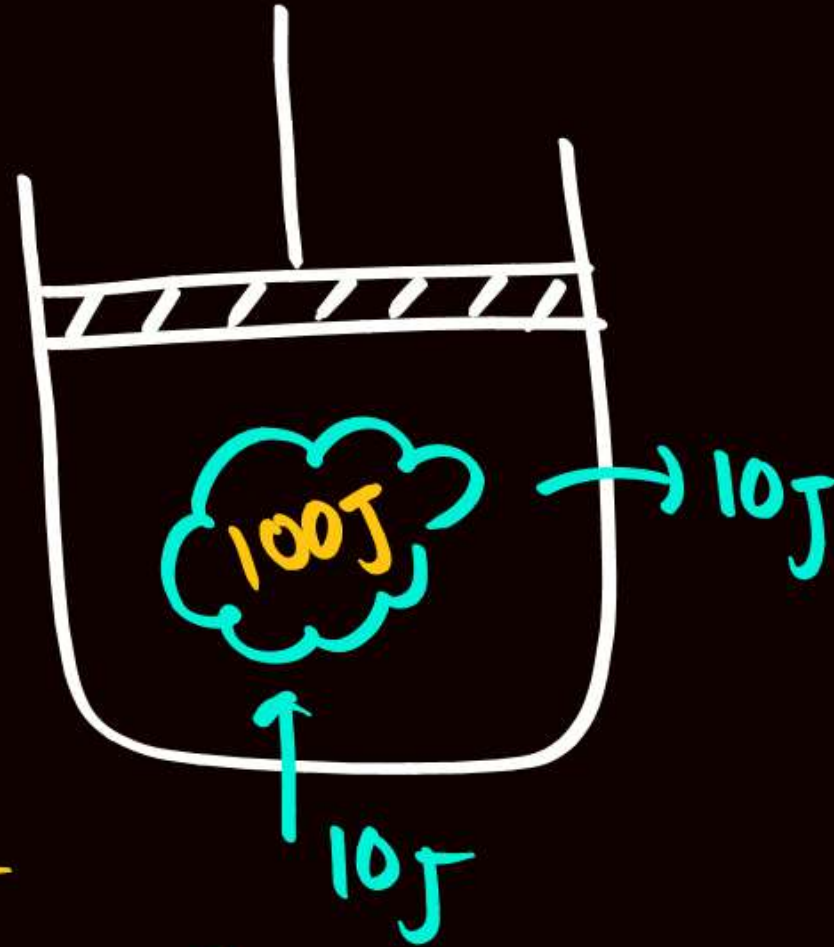
(Iv) Adiabatic process

⇒ The process in which heat of system remain constant

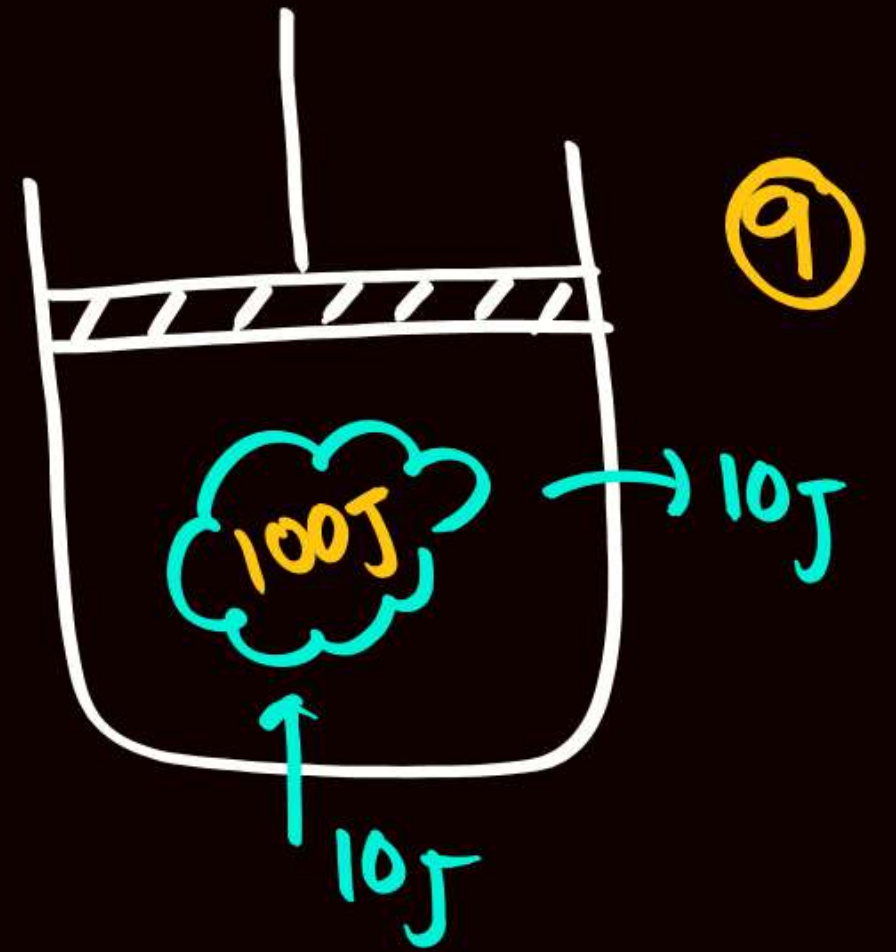
$$Q_i = Q_f$$

$$\begin{aligned}\Delta Q &= Q_f - Q_i \\ &= 100\text{J} - 100\text{J} \\ &= 0\end{aligned}$$

$$\begin{aligned}\Delta \theta &= 0 \\ d\theta &= 0\end{aligned}$$



$$Q_i = 100\text{J}$$



$$Q_f = 100\text{J}$$

Reversible Process

the process which can be reversed.

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Irreversible Process

the process which cannot be reversed.



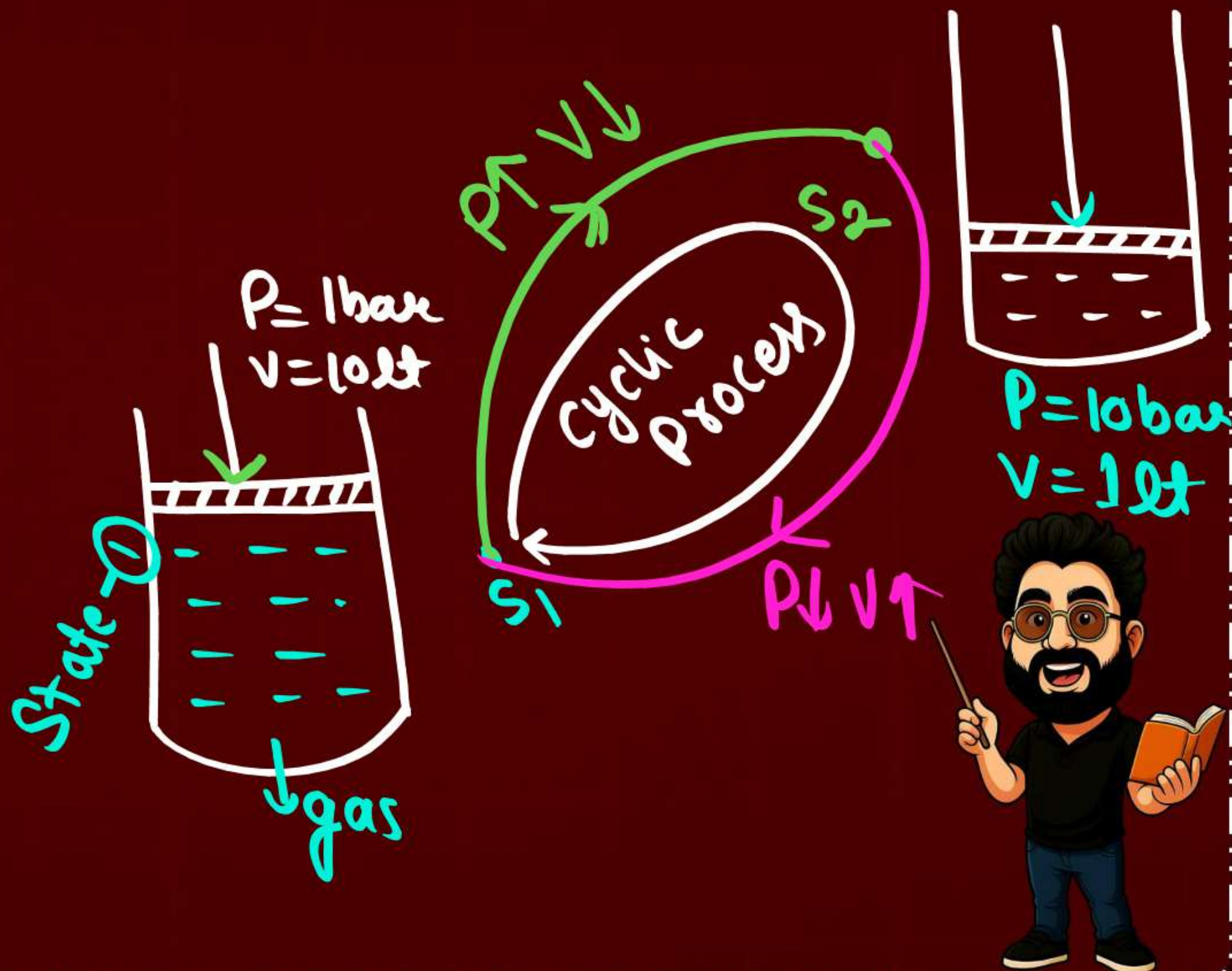
Cyclic Process

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PW

State-II

#The process in which state of gas change from one point to another point and Return to Same point



Temperature

- The degree of **Hotness** or **coolness**

K, °C, F

⇒ Heat travel from high temp to low temp.

100°C $\xrightarrow{\text{heat}}$ 10°C

10°C $\xleftarrow{\text{heat}}$ 100°C

50°C $\xrightarrow{\text{net Heat}}$ 50°C
do not travel

(Thermal equilibrium)

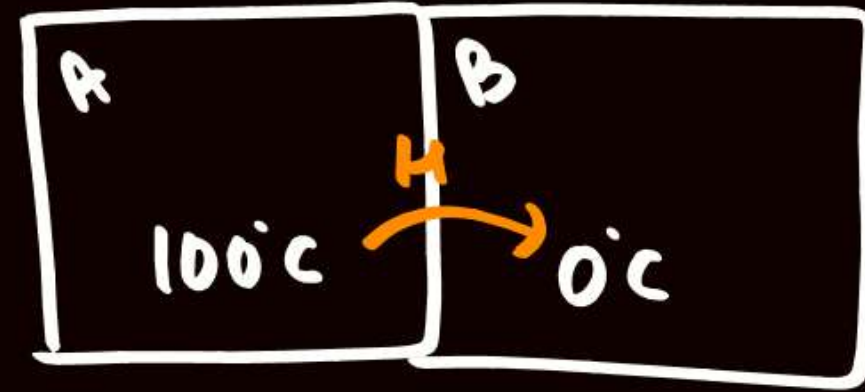


Thermal equilibrium

⇒ The Temperature of two body become Same

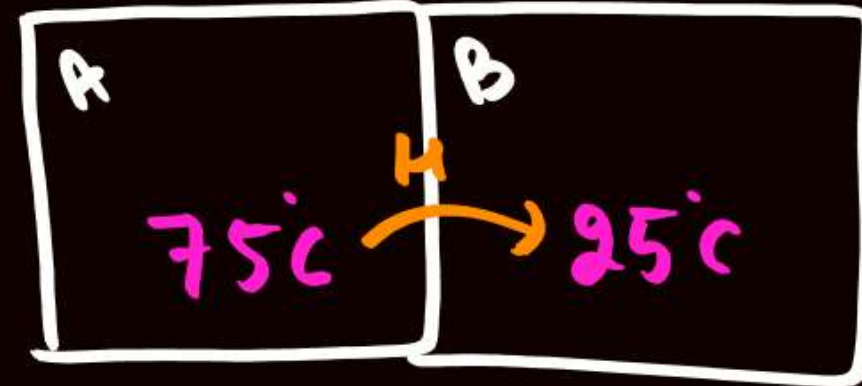
⇒ no net heat transfer

Case-I

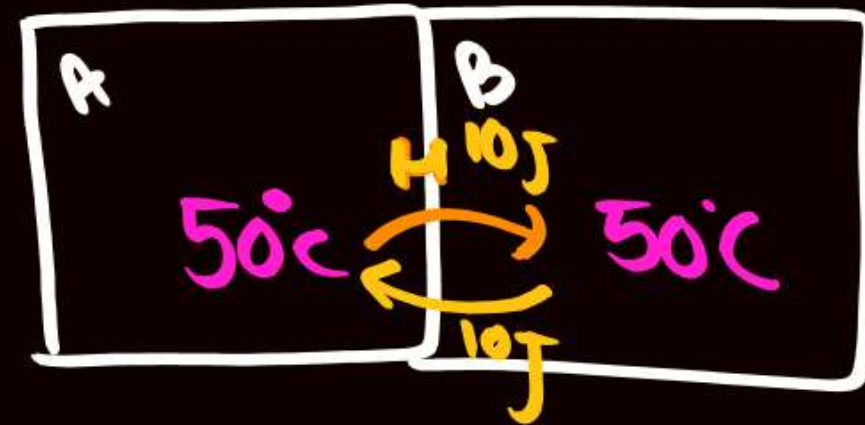


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Case-II



Case-III

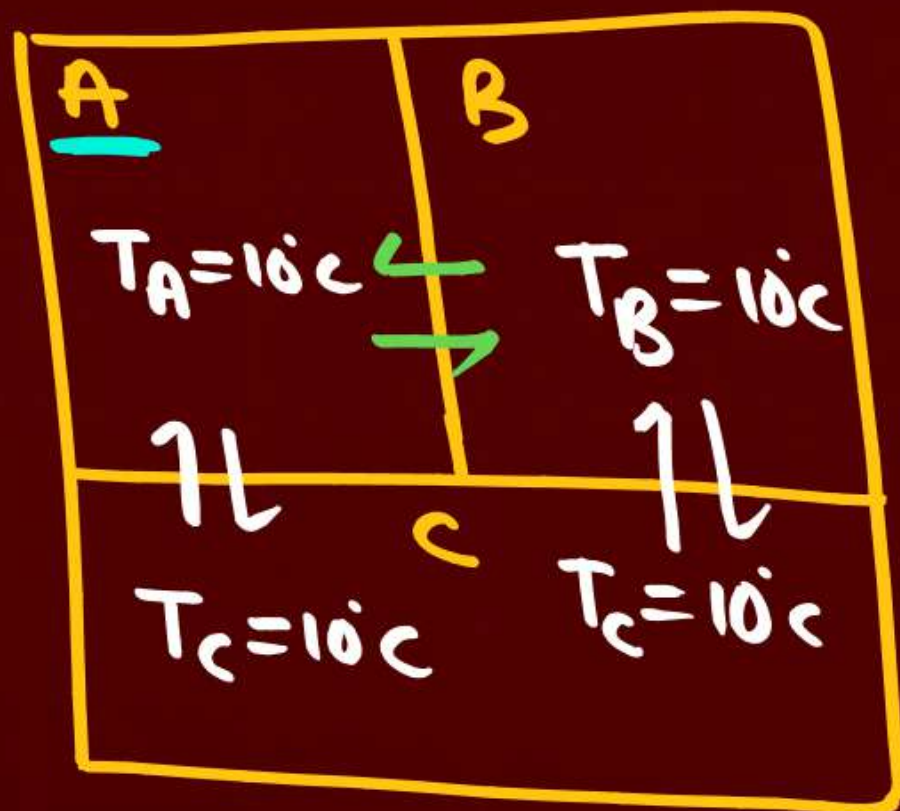


Zeroth Law of Thermodynamic

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If the temperature A is equal to temperature of B and temperature of $T_C = T_B$
Then Temperature $T_A =$ Temperature T_B



$$\begin{aligned} T_A &= T_C \quad \text{--- (I)} \\ T_B &= T_C \quad \text{--- (II)} \end{aligned}$$

from equation (I) and (II)

$$T_A = T_B$$



SUMMARY



→ TD

→ S

→ S

→ $U = S + S$

→ IT

IB

IC

A

HOMework



Q who give the zeroth law of thermodynamic?

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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